Home work 9

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Question 1

Create a regex that extracts quoted phrases, including the quotation marks. If there are multiple quoted phrases, they should be treated as separate matches. Test it on the following two strings.

```
# Test strings
albert <- "Einstein said, \"God does not play dice with the universe.\""
news_clip <- "The death toll makes the wildfire the deadliest since 1918,
when 453 people were killed in Minnesota and Wisconsin by the Cloquet & Moose
Lake Fires, according to the National Fire Protection Association. Losses are
estimated to be approaching $6 billion, after roughly 2,200 structures were
destroyed in West Maui across the 2,170 acres burned by the blaze. \"This is
the largest natural disaster we've ever experienced,\" Green said. \"It's
also going to be a natural disaster that takes an incredible amount of time
to recover from.\" The death toll makes the wildfire the deadliest since 1918,
when 453 people were killed in Minnesota and Wisconsin by the Cloquet & Moose
Lake Fires, according to the National Fire Protection Association. Losses are
estimated to be approaching $6 billion, after roughly 2,200 structures were
destroyed in West Maui across the 2,170 acres burned by the blaze."

# Extract quoted text including the quotes
regmatches(albert, gregexpr("\"[^\"]*\"", albert))[[1]]
```

```
## [1] "\"God does not play dice with the universe.\""
```

```
regmatches(news_clip, gregexpr("\"[^\"]*\"", news_clip))[[1]]
```

```
## [1] "\"This is\nthe largest natural disaster we've ever experienced,\""
## [2] "\"It's\nalso going to be a natural disaster that takes an incredible amount of time\nto recover
```

Question 2:

Create a regex that extracts email addresses in the _______@_______.____ format. The email address will be composed of letters or numbers. Test it on the following character vector.

```r
possible_emails <- c(
  "My email is scarden@georgiasouthern.edu.",
  "My twitter handle is @swcarden.",
  "Order your buds from four20@blazeit.com!"
)

# Extract emails
matches <- regmatches(possible_emails, gregexpr("\\b[a-zA-Z0-9]+@[a-zA-Z0-9]+\\.[a-zA-Z0-9]+\\b", possil
matches
```

```
## [[1]]
## [1] "scarden@georgiasouthern.edu"
##
## [[2]]
## character(0)
##
## [[3]]
## [1] "four20@blazeit.com"
```

Question 3:

Create a regex that can identify dates in a "mm/dd/yyyy" format. The connector can be either forward slashes or dashes, but can't be mixed. Just focus on the format; it's okay if it recognizes dates that are numerically impossible (01/32/2023).

```r
library(stringr)

possible_dates <- c("I was born on 09/19/1985.",
                    "Your arithmetic problem is to evaluate 12-12/1500.",
                    "01-01-2000 was the first day of the millenia.",
                    "Galileo's birthday was 02/15/1564. I forgot his present!",
                    "5/5/5555 isn't in standard format.")

str_extract_all(possible_dates, "(?:\\d{2}/\\d{2}/\\d{4}|\\d{2}-\\d{2}-\\d{4})")
```

```
## [[1]]
## [1] "09/19/1985"
##
## [[2]]
## character(0)
##
## [[3]]
## [1] "01-01-2000"
##
## [[4]]
## [1] "02/15/1564"
##
## [[5]]
## character(0)
```

4) Import the `faculty.csv` file we created in Lecture 8. Use a regex to extract the date from the `Terminal Degree` column and place into a new column called `Degree Date`. (5) (Grad student only)

```
library(dplyr)

faculty <- read.csv("faculty.csv")
faculty <- faculty %>%
  mutate(`Degree Date` = str_extract(`Terminal.Degree`, "\\d{4}")) %>%
  rename(`Office/Phone` =
           'Office.Phone...1.912..A....Armstrong.L....Liberty.S....Statesboro') %>%
  mutate(`Phone` = str_extract(`Office/Phone`, "\\d{3}-\\d{4}")) %>%
  mutate(`Office` = str_extract(`Office/Phone`, "[0-9a-zA-Z]+( [0-9a-zA-Z]+)?")) %>%
  mutate(`Degree` = str_extract
         (`Title`, "(M\\.?S\\.?|M\\.Phil|M\\.Ed\\.?|Ed\\.?S\\.?|Ed\\.?D\\.?|Ph\\.?D\\.?)")) %>%
  mutate(`Alma mater` = str_extract(`Title`, "\\(([^)]*University[^)]*)\\)")) %>%
  mutate(Title = str_replace(Title, " of Mathematics", "")) %>%
  mutate(Title = word(Title, 1, 2))


#knitr::kable(faculty) #table does not fit into PDF width, so not showing.
write.csv(faculty, "faculty_cleaned.csv", row.names = FALSE)
```